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Exploring the unmet needs for creating an enabling environment for nurturing care to promote migrant child health in Bishkek, Kyrgyzstan: A theory-guided community-based participatory action research

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ABSTRACT

Empirical evidence on community-driven child health promotion programs in disadvantaged migrant populations is limited despite various promotional strategies. Therefore, we implemented a developmental process to shape child health interventions using theory-guided community-based participatory action research (CBPAR) in a migrant community in Bishkek, Kyrgyzstan between 2015 and 2019.

The collaborative and iterative CBPAR process was conducted through participatory workshops and focus group discussions with support from trusting and collegial partnerships between community members and the research team. The goal and scope of the intervention was guided by enabling environments for nurturing care, including the four domains of caregivers' capabilities, empowered communities, supportive services, and enabling policies. Diverse interests and needs identified by community members were aggregated in the theoretical model and reflected in the intervention.

Community-driven intervention is perceived as a culturally acceptable, sustainable, sensitive and relevant approach to solve problems. There are several challenges in conducting the CBPAR, such as the effort and time spent on building partnerships, co-learning and mutual understanding, and the power equilibrium involved. Despite this, the success of the CBPAR process provided opportunities for community mobilization, empowerment and sustainability of the intervention. Evaluation of the process and outcomes of the intervention provided community health researchers and practitioners with evidence of the theory-guided community participatory approach.

1. Introduction

Migration refers to the large-scale movement of human beings or animals from one place to another in order to settle. This could be across an international border or within their own country (International Organization for Migration, 2019). It is currently estimated that there are around 272 million international migrants and an estimated 740 million internal migrants worldwide (International Organization for Migration, 2019). The complexity of migration has increased due to various factors including the migrant's legal status (e.g., irregular migration), voluntariness (e.g., forced migration or displacement, return migration), reasons for the movement (e.g., environmental migration, family migration, international students, labor migration and urbanization) and length of their stay. Although there is greater awareness on migration and its complexity across the world, the influence of migration on health remains poorly understood. Actions taken on behalf of the migrants' health protection and promotion also remain limited (Wickramage, Vearey, Zwi, Robinson, & Knipper, 2018). Furthermore, there is far less information on migratory movements within low and middle-income countries as compared to high political attention given to cross-border movements occurring in high-income countries.

Kyrgyzstan is a mountainous, landlocked lower middle-income country in Central Asia, with a population of 6.3 million and gross national income of $ 1281 per capita World Bank, 2019). The country is a relatively young state, with 32.4 % of the population under 14 years of age and life expectancy at birth estimated at 71.2 years in 2017 (75.4 years for females and 67.2 years for males according to a World Bank Group survey in 2019). It relies on a small economy dominated by minerals extraction and agriculture, which contributes to 36 % of the country's wealth. Although Kyrgyzstan was in deep economic recession for years after gaining independence from the Soviet Union in 1991, the

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country has taken large strides in economic development through major restructuring processes, demonstrating an annual GDP growth rate average of 3.38 % from 1994 until 2019 (The World Bank, 2019).

Kyrgyzstan is a country that has seen huge cross-border migration. The Ministry of Labour, Migration and Youth of the Kyrgyz Republic estimated almost 20 % of the Kyrgyz population work outside the country (600,000–700,000 Kyrgyz citizens), and remittances from migrant workers accounted for 30.3 % of Kyrgyzstan's GDP (The ministry of Labour, 2015). Economic and political factors are the main reasons for this large-scale external migration. The limited capacity of the domestic labor market, low employment opportunities, lower quality of jobs, and poor financial returns are some of the reasons behind migration (Ibraeva & Ablezova, 2016). Internally, active political and economic reforming led to extremely uneven regional development and drastic process of urbanization in metropolitan areas, whereas it resulted in severe economic downturn in the rural areas. This is because traditional economic activities such as the collective farming system had collapsed (Batsaikhan & Dabrowski, 2017; Daneykin, Andreevsky, Rogozhin, & Sernetsky, 2015). As political and socio-economic opportunities concentrated in urban areas, there was a sharp decline in economic opportunities in rural areas and migrants from rural areas moved to urban lands in search of better living conditions (Collinson, 2010; Ibraeva & Ablezova, 2016). It has been reported that from 1999 to 2009, approximately 30,000–40,000 people have moved within Kyrgyzstan annually (Ibraeva & Ablezova, 2016), which is nearly four times the number of international migrants (Hatcher & Balybaeva, 2013). During the same period, the population of Bishkek grew by 1,784,000 and as a result became increasingly divided by indigenous urban citizens and migrants (Ibraeva & Ablezova, 2016). Bishkek is now the most populated city in the country with internal migrants and they reportedly make up almost half of the total population in Bishkek.

Such excessively large volume of internal migrants increased the stress on urban centers, leading to an imbalance in population distribution, irregular urbanization processes and higher unemployment levels (Yüksel, Eroğlu, & Ozsari, 2016). Unlike the strong migrant integration policies held by European countries for many years now, Kyrgyz migrant policies, like the grounding law, were only recently introduced. Until internal migration began, Kyrgyz law prohibited dual registration for houses (Hatcher, 2015). This resulted in marginalization and discrimination of the migrants, which eventually will lead to inequalities in educational and employment opportunities, poor living conditions, and fewer benefits for the migrants from the public services in their new place of settlement (Agadjanian & Gorina, 2013; Yüksel et al., 2016). The unstable economic conditions, and dearth of public infrastructure and social support networks negatively impacts the health of the migrants (Shao et al., 2018) leading to regional health inequalities (Hatcher, 2017).

Children are the most vulnerable group affected by migration, with limited access to education and health care, which puts them at risk of poorer physical and psychological health outcomes (Issaka, Agho, & Renzaho, 2016; Markkula et al., 2018; UNICEF, 2016). Although the spectrum of child health promotion strategies to date in the vulnerable child population group (including SPRING project (Strengthening Partnerships, Results, and Innovations in Nutrition Globally), Gulazyk (Sprinkles) Micronutrient powder home fortification program, or ECD (Early Childhood Development) program, is diverse (Kan, Choi, & Davis, 2016; Zeng et al., 2018), the evidence on child health promotion programs in this population is limited. This can be attributed to a lack of contextual understanding of migrant children's health, i.e., the impact of migration on child health outcomes and socio-cultural barriers specific to migrant communities (Madhavan, Schatz, Clark, & Collinson, 2012; Omariba & Boyle, 2010; Wang, Su, & Wen, 2011).

In this regard, in 2015, we established a five-year stepwise community-based child health promotion initiative, "Bright Kyrgyzstan", supported by the Korean International Cooperation Agency (KOICA) (2015, 2016, 2017–2019) (Grant No: 2015-3, 2015-34, 2017-047), to improve child health and wellbeing in underserved migrant areas of Bishkek in Kyrgyzstan. The initiative included - A) a one-year community health needs assessment as part of the project incubating period (2015), B) a one-year pilot project providing proactive primary healthcare service focusing on child anemia prevention and oral health (2016), and C) a three-year community-driven child health promotion program (2017–2019). On the first year of the initiative, we conducted an analysis to identify community-specific health issues and the determinants of health. The most prioritized target areas for child health promotion were identified as follows: a) capacity building of health care providers (HCPs) b) proactive and preventive primary health care services c) mobilization of community resources d) improvement of community environment (including hygiene and safety) e) self-management capacity building for child health and well-being and f) the inclusion of a multidimensional approach to healthcare (Shin, Lee, Lee, & Shon, 2019). The pilot project for proactive community health care services was followed by community analysis in the following year (2016), with a mission to i) grow the clinical competency of HCPs ii) strengthen community capacity for child health management and iii) improve the quality of child health care services.

During this period, community members engaged in project activities to challenge community-specific barriers to ensure better access to a higher quality of primary child health services. One significant outcome achieved during the first two years of the initiative was ensuring community members' interest and expectations from the program. The initiative then advanced to the last phase of the community-driven child health promotion program. An open communication model and a collaborative relationship ensured the community was involved in all stages of the developmental process such as identifying community needs, designing the intervention, and its implementation and evaluation.

Therefore, the aim of this paper is to describe the theory-guided community-based participatory research (CBPAR) for developing a community-driven child health promotion program. This paper is organized as follows: the next section provides more information on theoretical backgrounds, including nurturing care and CBPAR; the third section informs on the material and methods such as study population and procedures; the fourth section presents the study results in four domains of nurturing care framework; and the final section will provide discussions along with lessons learned from the project and conclusions.

2. Theoretical backgrounds

2.1. Nurturing care

The ecological model of an enabling environment for nurturing care was adopted in this project to guide the developmental process (Fig. 1). Nurturing care refers to conditions created by public polices, programs and services that enable communities and caregivers to ensure good health and development in children (World Health Organization, United Nations Children's Fund, & World Bank Group, 2018). It comprises five components: good health, adequate nutrition, security and safety, responsive caregiving, and opportunities for learning. Nurturing care is especially important for children at the potential and substantial risk of physical, social, emotional, psychological, and political vulnerability, since child health and development are significantly conditional on these environmental factors. The ecological model of nurturing care highlights a "whole-of-society" approach to supporting families and caregivers in providing nurturing care for children. This is facilitated by enabling environments provided by policies, services, community, and family. The model shows the four domains of an enabling environment for nurturing care namely - caregivers' capabilities, empowered communities, supportive services, and enabling policies. Multi-sectorial coordination and community participation are core strategies for community-based health promotion projects. This needs an ecological approach to community health issues and related factors. This model

Fig. 1. Ecological model of an enabling environment for nurturing care.
Note: Adopted from "Nurturing care for early childhood development: a framework for helping children survive and thrive to transform health and human potential," (World Health Organization et al., 2018)

provides constructs for the ecological assessment of community needs and interests in providing nurturing care, which are reflected in the development of intervention components and implementation strategies.

2.2. Participatory Action Research (PAR)

PAR refers to a collaborative research method for change in action using iterative reflective action cycle, including data collection, reflection, and action (Baum, MacDougall, & Smith, 2006). Community-based PAR has been widely applied in deprived communities to identify and intervene in socio-structural barriers to potential and substantial unmet needs of the population's health status and quality of life (Phelan, Link, & Tehranifar, 2010; Winter & Munn-Giddings, 2001). It particularly stresses mutual relationship, equal cooperation, co-learning, empowerment, and critical reflection in the process of research design, including planning, implementing, and evaluating the community actions (Cargo & Mercer, 2008; Minkler & Wallerstein, 2008). It may contribute to improve the quality of research based on trustworthy relationship between researcher and community members. This will promote mutual respect and trust, facilitating greater participation in research (McDonald & Raymaker, 2016). Hence, PAR could create research process and resources that are more accessible and available to achieve community needs (Nicolaidis et al., 2015). The process of PAR may lead to community empowerment, skills and resources development, providing opportunities for involvement in decision-making experience, which ultimately fosters capacity-building for control over their lives (Paradiso de Sayu & Chanmugam, 2016; Zimmerman, 2000). PAR works in public health has been expanded to focus on changing health policies and health environments as well as individual's health behaviors, strategically diversifying their methods and approaches (Israel et al., 2010; Minkler, Garcia, Rubin, & Wallerstein, 2012). This approach would be more effective to practice the aforementioned ecological conceptual framework. This will comprehensively address unmet needs of child health promotion at individual, organizational and policy levels.

3. Materials and methods

3.1. Study population

The migrants lived in the outskirts of Bishkek due to lack of housing facilities in the central area. Such settlements are referred to as "novostroikas" in Russian (meaning "new settlement"). The degree of infrastructural development varied among "novostroikas," but it usually denoted a particularly underserved area where official urban planning and basic infrastructure was not available. Currently, only 48 "novostroikas" are officially recognized by the state authorities. The others remain illegal since the government is still reluctant to acknowledge their legal status. The authorities are fearful that developing infrastructure in these areas will lay unnecessary strain on the state budgets and resources to provide social protection and welfare support for this new population (Hatcher & Balybaeva, 2013; В каких жилмассивах Бишкека есть школы, 2017). Nearly 80 % of internal migrants in Bishkek did not have residence registration and were denied social benefits such as medical care, housing support, and education for their children (Hatcher, 2017). Although there is no officially reliable data demonstrating the migrants' accessibility to public health facilities, a gray literature that reported survey results conducted by an international NGO in 11 residential areas of Bishkek showed more than 30 % of the migrants aged over 16 years did not have access to free public healthcare. They had to instead make full or partial payment or use private medical services due to their illegal resident status or absence of registration at Family Group Practices (FGP) (Bermet Zhumakadyr kyzy, 2012). The study also revealed that at least 44 % of migrant children under 16 years had limited access to public health services due to the absence of a birth certificate.

The largest "novostroikas" are Ak-Orgo, Ak-Ordo, Archa-Beshik, and Kok-Zhar. Our sample communities were Ak-Ordo and Ak-Orgo in Lenin district, located on the south-western edge of Bishkek. These communities were chosen due to their greater involvement and expectations of change, which we gathered during the initial phases of our initiative. The success and sustainability of community programs depend on the engagement and commitment of the community in the activities. Both Ak-Orgo, formed in 1989, and Ak-Ordo, recently formed in 2005, consist of eight villages each. Of the total population, approximately 30 % of adults aged over 18 years were unregistered on the resident registration system (Shin, 2015).

The educational and health infrastructure was not enough to meet community demands. Ak-Orgo had four private kindergartens and two primary schools №77 & 84, while the only primary school №96 in Ak-Ordo was constructed at the end of 2018. In response to overcrowding in schools, the schools adopted a triple-shift system. Regarding primary healthcare, family group practices FGPs were the front-line providers of public health care services including general consultations, immunizations, and basic laboratory tests. A total of 35,994 people lived in these two regions with about 5000 children under five years. №1 Family Medical Center (2019). In total, eight family doctors and eleven nurses worked at the FGP in Ak-Ordo, and nine doctors and thirteen nurses at the one in Ak-Orgo. While the Den Sooluk national health care reform policy recommended one FGP per 1500 people within a median distance of one to two kilometers or distance of less than 30 min of travel time to the nearest health facility, only one FGP was located in each target area. According to a World Bank study (2019), the nurse-to-people ratio (per 1,000 people) in Kyrgyz was 6.3 compared to 8.5 in Kazakhstan, 12.1 in Uzbekistan and 7.0 in Europe & Central Asia (excluding high income countries). The doctor-to-people ratio (per 1,000 people) was 1.9 in Kyrgyz compared with 3.3 in Kazakhstan, 2.4 in Uzbekistan and 3.0 in Europe & Central Asia (excluding high income countries). The nurse-to-people ratio was 24 per 35,994 (equals to 0.67 per 1,000) and the doctor-to-people ratio was 17 per 35,994 (equals to 0.47 per 1,000) in the two migrant regions. The disproportionate allocation of resources in this area compared to the national ratios aggravated regional health inequities and negatively impacted community health.

Table 1
Community-specific interests and needs identified from various community member groups.

Domains of enabling environment for nurturing care	Academic researchers	Collaborating organizations	Health policy makers/ administrators	Primary HCPs	Mothers*
Caregivers' capabilities					● Personal knowledge and skills for home-care practice
Empowered communities	● Expansion of nursing paradigm: from disease-focused nursing practice to community nursing practice	● Community resource networking ● Strengthening community action	● Empowering community-based organizations ● Utilization of existing community leaders	● Strengthening connections between health systems and community ● Nr-CHW coordination ● Empowering village women as lay health workforce ● Safe and good-quality environment: environmental improvement of health care center	● Nr-CHW communication ● Empowering village women as lay health workforce ● Safe, accessible, high-quality child care facility ● Raising awareness of public health program
Supportive services	● Research networks for linking scientific evidence to clinical practices	● Improvement of communication flow among health care delivery systems		● Decision support with Evidence-based guidelines in daily clinical practice ● Interpersonal communication skills ● Networking with other community resources ● Service coordination (Nurse-CHWs, inter-health care delivery systems) ● Effective and efficient communication channels between Nurse-CHW ● Equipped for community child health care	● Increase in family contact with health system ● Effective and efficient communication channel between Nurse-CHW
Enabling policies			Actionable updates on activities to materialize "*Den Sooluk*"	Actionable framework for planning community activities in the primary health care setting	Affordability of basic health services; Expanding health coverage

* Those involved in community activities as community health workers (CHW) in the previous phase of the initiative.

3.2. Study procedure

3.2.1. Expanding relationships with community partners

During the first year of the initiative (the project incubation period), we conducted situation analyses to identify contextual health determinants and unmet needs of child health promotion within the community. We navigated three villages, one community center, three public health centers and local clinics, three primary schools, and governmental and non-governmental organizations to cultivate community relationships and networks with various levels of stakeholders. The forty-five community members we interviewed included researchers, policymakers, administrators, HCPs, non-governmental community activists and community mothers. They informed us of community-specific issues and the problems of child health management from different viewpoints. More detailed information of situation analysis is available in a different study. (Shin et al., 2019). In the second phase of the initiative, a pilot project was conducted to enhance proactive healthcare services, which considerably contributed to raising awareness of child health promotion in the community and mobilizing stakeholders as key partners. Existing community relations were expanded and reorganized to establish community-based organizations including steering and advisory committees and a community advocacy group.

3.2.2. Collaborative and iterative CBPAR process

The collaborative and iterative CBPAR process included multiple contacts with community members through various communication channels. To ensure evidence-based decision-making and reduce the gaps between community members' interest and the scope of the program, the cyclic process moved back and forth between the community context and extensive literature.

A series of participatory workshops was held with various community health stakeholders including mothers living in the target community, HCPs, local academic researchers (especially, from Kyrgyz State Medical Academy, KSMA), health policymakers and administrators, and local and international NGOs engaged in community health promotion and development. In the early stage, the workshops were mainly informative, sharing the activities and achievements of the past two years and the future challenges and scope through the remaining period of the initiative. The sessions also provided more information on the concepts of CBPAR and nurturing care was provided for the audience. Subsequent workshops were customized for the community members. Focus group discussions were held at the end of the workshops to discuss the agenda of each workshop. Six focus groups were created, which consisted of academic researchers, collaborating organizations, health policymakers and administrators, primary HCPs, and representatives among mothers. Discussions were guided by a theoretical framework, which was helpful in itemizing the challenges and opportunities the community faced and developing responsive strategies.

We also conducted individual in-depth interviews with participants who wanted to discuss the agenda after the formal focus group sessions. The one-on-one format helped us probe issues to obtain further insight. Individual interviews were conducted with community parents on several informal occasions. Those engaged in community activities from the beginning of the initiative mainly led these interactive sessions, providing health service information and asking people about their opinions (including their feelings, attitudes, beliefs, and experiences) on the enabling environment for nurturing care.

The individual interactions helped us get in getting more ideas on intervention strategies (content and operating methods, target a person in charge of the suggested activity, and make resources and networks available to support the activity). The preliminary intervention design was reviewed in the community-based organizations, especially the advisory committee. To adhere to the rigor of CBPAR principle, contextual reflection via communication and co-learning strategies was maintained throughout all stages of the project. The feedback obtained from the community was useful to customize the activities according to the scope of the program.

The situation analysis described in this study was conducted during the first three months, from January 2017 to March 2017, when the third phase of the initiative started. But adhering to the CBPAR principle, the iterative cyclic process is still ongoing and reflects in the project management cycle.

4. Results

4.1. Identified community-specific interests and need for an enabling environment for nurturing care

The factors required for an enabling environment for nurturing care were identified after discussions with community members (Table 1). The contents in each domain of the ecological model for nurturing care were interconnected and interdependent.

4.1.1. Determining caregivers' capabilities

Mothers were most in need of support in home-care practices. They addressed that there is a lack of skills and insufficient support for child health management. According to a lady in the community, *"I usually ask my mother or other elderly women in the village about all the doubts I have during pregnancy. However, sometimes it is not enough. I wish I had a professional person from whom I could get information or learn best practices for child rearing."* It was found that some common home-care practices for children were undesirable. For instance, giving traditional tea as a drink instead of fresh water; preparing baby formula with caffeinated tea etc. There were zero opportunities to learn new skills in self-care. This implied the need for a communication strategy to strengthen home-care and address unsafe practices that already prevailed in the community.

4.1.2. Empowering communities

All groups of community members strongly agreed with the need to strengthen capabilities across all aspects of healthcare including community partnerships and resource mobilization. A mother who served as a community activist said, *"I inform parents about public health programs that they can benefit from. For example, an anemia test is free of charge for children under the age of five, and when a child is diagnosed with anemia, the parents can get a voucher for iron supplements. However, parents are reluctant to take the test, and some skip the scheduled child health examination although I remind them. I think it is because they do not consider it important. They think everything is alright because their baby is not sick."* Other mothers agreed on the significance of raising awareness and informing families about being smart consumers of health services.

During the interviews, representatives of international NGOs such as UNICEF in Kyrgyzstan, addressed the significance of community resource networking and engagement in community action. Health policymakers and administrators suggested empowering existing leadership and local organizations as a resource utilization strategy. This was found to be effective in increasing awareness of health-related issues among community members. The HCPs working at family general practices mentioned the need to strengthen connections between health systems and community to support families. However, the current health workforce was unable to reach out to all corners of the community due to shortage of resources and staff. Discussions were also held on the most effective way to increase coverage of basic health services and efficient utilization of community health workers (CHWs). State-appointed village leaders named "Okz" represent their villages at the national level and have priority access to useful information and resources. Although they play a leader's role as agents of both the state and village community, they are not specified for health delivery services. The HCPs and mothers suggested recruiting village women as CHWs especially those who have an in-depth understanding and experience in childcare. Family nurses emphasized coordinating with the nurse-CHW, saying, "*We have too much to do in the office and not enough time to visit and follow-up on a patient's condition after their visit to the FGP. Women community activists in the projects helped lighten the burden of caring for patients in remote areas. They visited patients' homes and reported their condition to us. Based on their reports we trained them on what they need to know and the messages to be delivered in health education. We appreciate their activities.*" The discussions then focused on effective ways to qualify and utilize CHWs.

The discussion on this domain specifically inspired academic researchers, who also deliberated upon the need for capacity building for epidemiology and community health. Aware of the significance of nurses' role in community health promotion, they understood the need to expand the nursing paradigm from disease-focused to community nursing practice. "*So far, we only focused on teaching about disease management and emphasized the need for improvement in clinical skills. Our curriculum does not include a population health practicum.*"

Inadequate logistical facility at public health centers was also noted. Mothers with little children waited a long time, sometimes even outside the FGPs due to insufficient space. There were no private feeding room facilities for breast-feeding mothers, neither were there enough washrooms. The HCPs requested that the healthcare center be renovated, and the women proposed a childcare facility in the village.

4.1.3. Strengthening supportive services

Regarding strengthening the quality of services in the primary health care system, the HCPs wanted to increase the competency of daily clinical practice. They suggested the need for a decision support tool for evidence-based practice and better medical equipment. The distribution and allocation of HCPs were not regionally equitable, and their level of competency varied. The opportunity to build capacity through training programs was limited only to the HCPs working at FGPs. As for the pre-registration program for nurses, the curriculum mostly focused on disease-specific practices in clinical settings. The nursing practicum did not include population-based health promotion, and post-registration training programs on primary health care services were not available.

The HCPs also focused on the relationship between the health professionals in the system. Representatives from an international NGO agreed that communication flow in the health care delivery system needs to be improved, highlighting the hierarchical communication culture. They anticipated improvements if a horizontal communication network was in place, which would increase work efficiency and autonomy in the system. A family doctor who spoke about the challenges of communicating with authorities sitting higher up in the health care delivery said that referral process for high-risk patients needing further examination was always affected. This was attributed to delayed or no feedback from higher authorities. They also mentioned that their opinions or demands were sometimes overlooked and did not reflect in the corresponding action. They often felt pressured to do top-down tasks. The importance of communicating efficiently with patients was also looked into. A family nurse said, "*I sometimes feel like I do not interact enough with my patients, mothers, or other caregivers of young patients. After explaining the results of a medical examination and/or providing the required treatment services, I just say goodbye to them,*" Another nurse mentioned, "*What more should I ask the mother of the child? I already know what I need to know when I examine the baby, and I patiently listen to them and their questions. I admit I do not ask additional questions regarding self-care, or what they want to know regarding the treatment unless they ask me themselves. I rarely reflect on the patient's understanding of what I said, so I am sometimes not sure that the mother has understood what I said.*"

Mothers wanted better access to the health care system, and requested increased involvement. They welcomed the utilization of a CHW for home-visit services, and also expressed interest in serving the community as CHWs. A former member suggested establishing community outreach services, i.e., home-visit services, noting: "*I am often required to visit the health center after visiting a patient's home. Sometimes I feel too exhausted to walk the distance. If I have another scheduled visit, I cannot go back to the health center again. I end up forgetting some important patient information in my report. We need to use mobile phones and other technology to find a way to work around this.*" It was suggested that technology be adopted as a supportive tool which could improve the accuracy and effectiveness of home-visits and nurse-CHW communication.

And finally, the KSMA researchers suggested an association between research and health practices. They expected that networking between academic investigators and HCPs might contribute to building empirical evidence of community programs or services. The HCPs agreed to have collaborative relationships with academic researchers, expecting this would strengthen decisional support with evidence-based guidelines. Policymakers also mentioned that empirical evidence of community health services could contribute to realistic policymaking.

4.1.4. Enabling policies

Policymakers and the HCPs noted the role of *Den Sooluk* to achieve health equity. This is because legal mechanisms alone are not sufficient to allow universal coverage of all citizens or equal access for poor and vulnerable groups. For example, the state guaranteed benefit package (SGBP) provides primary and emergency care services free of charge to the entire population. It is also equally available for internal migrants enrolled in the FGP in their place of enrolment without a permanent residence permit. However, many internal migrants are unaware of this and do not clearly understand the rules. Patients from low-income groups had the right to a copayment exemption but ill-defined mechanisms ensured that they were unaware of this entitlement. There was also no clarity on the process of obtaining documents to certify one as low-income. Because of the limitations in the benefits that SGBP provides, people ended up paying for their medication and hospital expenses. Medical expenses for prescriptions were exempted up to 50 % of the total expense. However, many people cannot afford their medication due to poverty. Regarding the unaffordability of basic health services to community residents, the HCPs highlighted the need for advocacy to raise awareness of public health programs. According to one nurse, "*The government provides medical subsidy that covers most medical expenses, however, the villagers feel burdened by out-of-pocket payments for health care. I think the people in this village feel less comfortable paying medical costs than those in other villages. Some parents refuse to go for further examinations because they cannot afford.*" Both health policymakers and HCPs highlighted the need to equip the *Den Sooluk* with achievable and affordable actions in the real world. For this, political and policy attention along with financial support and investment was proposed in the migrant villages.

4.2. Intervention components to create an enabling environment for nurturing care

Viable and pragmatic intervention components were created in each domain of the ecological model of nurturing care. The performance goal and scope were aligned with the mission of our initiative. From the ecological perspective, a complex intervention design included interdependent and intertwined multifaceted activities that had synergistic effects on one another. Each intervention component with its performance objectives, outcome measures, and evaluation methods are described in Table 2. Action plans for each intervention were reviewed during the multiple, cyclical discussion sessions, reflecting the feedback from community stakeholders in the revision and adjustment of the interventions.

4.2.1. Determining caregivers' capabilities

The mothers and HCPs suggested community health education programs to strengthen caregivers' capabilities. The topics they were interested in included anemia and child nutrition, parasite infections,

Table 2
Intervention design based on the ecological model of nurturing care.

Domains of an enabling environment for nurturing care	Intervention components	Performance objectives	Outcome measures	Evaluation methods
Caregivers' capabilities	Community health education program	- Increasing knowledge and skills for child health and wellness of community members - Strengthen self-confidence regarding child health self-management practice	- Completeness and fidelity of education session (process, outcome) - Adaptability and satisfaction (outcome) - KAP for healthy growth and development *(outcome)*	- Review of operating manuals and workbooks - Review of education program modules - Structured KAP questionnaire survey - Post-session evaluation
Empowered communities	Community resource mobilization	- Mobilizing social network linkages - Organizing community resources - Strengthening community actions	- Establish advisory and steering committee and Nurse-CHW meeting - Commitment of the members in each committee	- Review of meeting log of advisory and steering committee and Nurse-CHW meeting
	Community advocacy	- Increasing community awareness - Creating a safe, stimulating, and supportive community environment	- Community campaign (health fair, health contest, on-the-spot health screening events, family walking day, village outdoor cinema, etc.) (outcome) - Community engagement (outcome, impact) - Community cohesion and autonomy (outcome, impact) - Availability of networks (outcome, impact*)*	- Participation in community events - Communication log for development of community events - Focus group discussion and individual interviews - Evaluation of community participation using Spider-gram
	Environmental improvement	- Creating a safe and high-quality environment - Improving environmental hygiene	- Environmental renovation (plumbing, toilets, electricity, heating and cooling systems) in primary schools and FGPs (outcome) - Construction of child health promotion center *(outcome)*	- On-site inspections
Supportive services	Point-of-care protocol for primary child health care	- Encouraging decisional support for evidence-based practice - Enhancing the quality of primary health care service Increasing competency of clinical practice - Increasing patient safety - Re-orienting health services	- Completeness and fidelity of point-of-care protocol and on-the-job training module (process, outcome) - Adaptability and satisfaction (outcome) - Adherence to the protocol (process, outcome) - Satisfaction with the service of community residents (outcome, impact) - Organizational adoption of the program *(impact)*	- Review of communication and meeting log for point-of-care protocol - Review of operating manuals and workbooks - Peer and supervisory observation - Evaluation of clinical competency using LCJR - Service satisfaction survey - Organizational adoption of the program
	Home-visit service	- Increasing accessibility of the service and support - Empowering caregivers and families - Developing community resources by training and empowering CHWs	- Completeness and fidelity of CHW training module (process, outcome) - Adaptability and satisfaction (outcome) - KAP in CHWs (outcome) - Frequency of home-visit practice (outcome) - Compliance with the home-visit protocol (process) - Satisfaction with home-visit services of community residents (outcom*e)*	- Review of training module - Review of operating manuals and workbooks for home-visit practice - Review of communication log between nurses and CHWs - Peer and supervisory observation Service satisfaction survey
	Innovative communication support tool	- Developing information systems to facilitate nurse-CHW communication - Strengthening support in primary health care service	- Adaptability and satisfaction (outcome) - Usability of the tool (outcome) - Organizational adoption of the program (outcome, impact*)*	- Analysis of message log - Satisfaction survey of nurses and CHWs
	Capacity-building training program	- Improving HCPs' clinical competency - Enhancing communication flow among HCPs - Participatory problem-solving	- Completeness and fidelity of training module (process, outcome) - Competency in clinical practice (outcome) - Adaptability and satisfaction (outcome) - Perception and attitude of effective communication and mutual respect *(outcome)*	- Review of operating manuals and workbooks - Post-training satisfaction survey - Individual in-depth interviews
	Provision of medical devices & supplementary education materials	- Equipping the FGP for a better quality health care service - Providing relevant and accessible education materials	- Medical equipment and its supplements (outcome) - Printed education materials (outcome) - Usability of the materials *(outcome)*	- Review of operation manuals - Review of utilization log
				- Review of meeting log

(*continued on next page*)

Table 2 (*continued*)

Domains of an enabling environment for nurturing care	Intervention components	Performance objectives	Outcome measures	Evaluation methods
	Research networks for linking scientific evidence to clinical practices; capacity-building in scientific skills and research training	- Creating and enhancing network linkages between the research academy and community health field - Developing community nursing practicum - Developing and implementing collaborative research work - Capacity-building for scientific skills and research training for KSMA researchers	- Establishment of taskforce for development of community nursing practicum, consisting of the members from the FMC and FGP (outcome) - Completeness and fidelity of CHW training module (process) - Research proposals and articles (*outcome*)	- Curriculum of community nursing practicum Research proposals and articles
Enabling policies	Monitoring, evaluation and dissemination of results	- Corresponding to the national health care reform policy – feedback to policy makers - Providing examples of success regarding actionable updates on the policy and administration	- Memorandum of Understanding (outcome) - Community forum (outcome) - Organizational diagnosis and feedback (impact) - Political and policy attention (*impact*)	- Community engagement in the project (activeness of advisory committee) - Communication logs in community forum - Organizational adoption of the program

CHW: Community health worker; FGP: Family group practice; FMC: Family medicine center; KAP: Knowledge, attitudes and practice.

sanitation, parent-child interaction, good parenting, and so on. The training-of-trainers (ToT) model was adopted as an educational strategy for community health education. A Korean health expert trained the HCPs to build a pool of competent instructors who can coach other HCPs and CHWs. Trained CHWs visited homes to monitor the health of the children and deliver health information according to the home-visit protocol. The HCP and CHW training sessions were held every two months. In addition to the regular sessions, additional sessions were held when necessary.

4.2.2. *Empowering communities*

The community empowerment programs included community resource mobilization, community advocacy, and facility renovation. Community members were organized according to their functionality and productivity, and an advisory committee, steering committee, and community advocacy group were established. The newly established community-based organizations were operated based on autonomous internal rules enacted through the consensus of the members of each organization. They autonomously and collaboratively developed community advocacy activities such as health fairs, health contests, a family walking day, children's day festival, and movie nights. They believed these activities would increase community awareness of child health promotion and create a safe and supportive environment for childcare. Facility repair and renovation was implemented in cooperation with these community-based organizations. Community child health promotion centers were newly constructed in both the FGPs at the request of community members

4.2.2.1. Organizing advisory and steering committees. An advisory committee was organized to advise the research team, enabling them to formulate appropriate interventions. It was agreed that their role in this project would be to provide technical support and peer review throughout the duration of the project, working in a cooperative and collegial manner to support the success of the project. They were actively involved in the problem-solving and decision-making process to resolve project challenges, and played a role in harnessing the network's ability to mobilize community resources. Member organizations of the advisory committee included the Health Promotion Center in the Ministry of Health, Bishkek Health Promotion Center, district Family Medicine Center (FMC) responsible for the FGPs located in its catchment area, FGPs in the AK-Ordo and AK-Orgo areas, UNICEF Kyrgyzstan, KSMA, District Office, District Education Office, and two regional primary schools. Ordinary advisory committee meetings were usually held twice a year. The steering committee was expanded to serve as project coordinator and the primary communication link between the research team and community members, which mainly consisted of HCPs from the FMC and FGPs as well as CHWs. This collaborative group met monthly and discussed practical issues pertaining to implementing the intervention, and provided feedback from a front-line perspective.

4.2.2.2. Organizing community advocacy groups. A mother's club for community advocacy was organized based on lessons learnt from the previous years of the initiative. Volunteers as community activists were primarily sought among existing community leadership networks, particularly traditional village leaders, the "Okzs". Since "Okzs" were primarily elderly males in the authoritative and hierarchical culture, who did not actively participate in childcare activities, they were less interested in nurturing care than other social activities. Therefore, we recruited women interested in childcare with relevant experience as community activists, and organized a mothers' club. Members were recruited through the extensive networks of community activists. The members who agreed to voluntarily engage in community health services, especially the home-visit service, were recruited as CHWs. Newly recruited mothers were empowered with health information and provided with skills through job training. Furthermore, they were affiliated with FGPs as lay CHWs to support the home-visit service on behalf of nurses. Although they did not receive a formal salary from the state or project, they were rewarded for their services with prepaid mobile Internet expenses and opportunities to participate in conferences, workshops and other health promotion events. In total, 14 women in Ak-Orgo and 30 in Ak-Ordo were trained, and served the community as CHWs. Nurse-CHW meetings were held quarterly, where CHW training sessions were provided and discussions regarding the practice of the home-visit service conducted.

4.2.3. Strengthening supportive services

Interventions were created to strengthen the primary child healthcare services, focusing on re-orienting primary healthcare services, capacity-building training programs for HCPs, equipping facilities with medical devices and supplements, and adopting communication support technology. In terms of re-orienting primary care services, a point-of-care protocol for preventive child health care services was developed and home-visit services activated. Using CHWs in the home-visit service aimed to increase family contact with useful health information and empower mothers. The mobile telecommunications service (MTS) was used in the home-visit service to guide follow-ups, refer patients, and for home-visit practices. Until then, nurses and CHWs had communicated through non-standardized, verbal systems. The nurses requested a follow-up visit for high-risk children and the CHWs reported their activities through face-to-face talks or phone calls. Now, using the MTS installed on their mobile phones, CHWs document their activities using a code number. The completed activity summary is transmitted to the central server for storage and to the nurses' PDA tablets in real-time. The MTS functions when cellular network coverage is available.

Capacity-building training programs encompassed up-to-date primary health care practices and communication skills for interactions between health professionals and patients. Various delivery strategies including participatory problem solving, supervisory shadowing, role-playing, and demonstration were adopted. Monthly meetings for HCP ToT training sessions were conducted. Medical devices and equipment for the physical assessment and diagnosis of anemia was requested.

Finally, a community health research taskforce team was established consisting of KSMA nursing scholars and HCPs in the FPGs. Research workshops were created to develop the research capacity of KSMA researchers, enabling them to be more responsive to public health needs.

4.2.4. Enabling policies

Evaluation strategies and dissemination plans were also included in the intervention planning to ensure the sustainability of community-wide health promotion policy changes. Considering the influence of the community-based child health promotion initiative at the organizational and policy levels, the mission and goals of the intervention were fundamentally reflected in the national health policy context. Health policymakers and administrators, and local and international advocacy organizations were involved in the project as members of the advisory committee. All study project procedures were approved by the Kyrgyz Ministry of Health and Institutional Review Board of the University (IRB No. KHSIRB-17-022), which also reviewed and approved all written material. As in the previous phase of the initiative, community data were collected via project activities, and progress was formally shared and discussed at the community forum and progress reporting sessions, in which health policymakers also participated.

5. Discussion and future research

The CBPAR process for intervention mapping is complex due to the consideration of the community needs in cultural and social contexts and is also challengeable to sustain community engagement and equitable partnership. Especially, in designing complex interventions which have more than one active component, the gaps between needs and capacity of the community could be greater than in a single component of intervention. Theoretical framework was particularly helpful to stick to the point of agenda which was used to address health-related issues of community in organized constructs.

Our efforts devoted to upholding iterative discussions and transparent communication could facilitate active community engagement through a collaborative process. However, one of the important facilitating factors was thought to be the attributes of the community and situational process, that is, motivation and capacity for reflection and group process of collaborative responses to challenge (Tse, Palakiko, Daniggelis, & Makahi, 2015). Our target community has been involved in community works of situational analysis and pilot project for 2 years prior to this project. Through participatory process during that time the community members acknowledged the purpose and value of the project and collaboratively responded to the challenges they encountered. It is observed that longtime relationship and mutual trust facilitated active engagement and accountability in the project. However, community participation or engagement is a sort of social process and situation specific so there is no gold standard to generalize this situational process in the other community setting (George, Mehra, Scott, & Sriram, 2015). It is essential to develop situation specific ways to ensure community participation in the early stage of the program.

The nurturing care framework includes the significance of creating a health-enhancing policy. Multiple actions in the community-driven intervention had key drivers to enhance potential political power in bringing about policy change. They were used to establish multilevel partnerships at the local, state, national, and global levels to enable coalition activities to acknowledge the mission of public health from the lower to higher levels of health organizations. Some community activists involved in the pilot project were officially appointed as "Okzs"—village leaders—and their role expanded from that to health navigator in the project to official community leader. Appointing women as village leaders is exceptional in Kyrgyzstan, a patriarchal society, since state-appointed village leaders are traditionally men. The attitudes and social norms governing women's role meant they carried out their traditional family roles at home. These norms did not support their active involvement in social activities (World Bank, 2011). Appointing women as official village leaders reflects women's empowerment in the local administration system, affording them greater privilege in accessing useful information from outside the village and enabling their participation in the modern bureaucracy. Considering the impact of community empowerment on political attention and engagement, we expect the success of the project to extend to achieving community-wide change.

CBPAR advocates active and equal partnerships aligned with open communication, mutual trust, co-learning, and power sharing. Despite the well-known benefits and strengths of involving community members, there were some challenges in conducting CBPAR in reality. The biggest challenge was that it is a time-consuming process, requiring access to key community representatives, cultivating and sustaining trusting partnerships and collaboration, and scheduling various meetings. In this study, we spent two years (2015–2016) building partnerships and capacity, collaboratively implementing the community analysis and pilot projects during the previous phases. At the end of the second year, we were assured of the collegiality and mutual trust of community partners. Community representatives became more engaged in the project, obtaining accountability and ownership, and discussions advanced to the future work in the next phase. In this stage, the community suggested their role in the work. Based on this partnership, community members' involvement was quantitatively and qualitatively expanded.

In addition, balancing power between community partners and the research team was another challenge in the CBPAR. Traditional community-based research in developing countries has been conducted "on" the community, positioning community partners as passive beneficiaries with little accountability (Otterbach et al., 2018; Tapp &

White, 2013). The research team kept in mind that a major challenge to the success of CBPAR is engaging community partners and empowering them with greater accountability to their community. At the start of the initiative, the research team took a leadership position, leading workshops and discussions and setting the agenda as starting points for the discussions. This was needed to inform community partners about the goal and process of the project and ensure productive and useful discussions. However, after repeated sessions, the leadership and contributions were gradually transferred to community partners. In some cases, the moderator in each committee and advocacy group led the meetings and the research team was involved as an outsider expert. This made participants feel more comfortable and open-minded in a relaxed atmosphere and enabled them to share their own feelings, opinions, attitudes, beliefs, and experiences. Community partners then primarily made decisions regarding implementation such as convening the meetings and workshops or education sessions, appointing trainers and recruiting participants, and planning episodic events. Power sharing or equal relationships between community partners with different interests is also crucial to community empowerment (Harris, Pensa, Redlich, Pisani, & Rosenthal, 2016; Tapp & White, 2013). Furthermore, engaging diverse community members in community-based organizations provided diverse perspectives on health needs and interests as well as different points to consider. It also provided opportunities to provide information to each other in various situations. Policymakers were informed of the reality of unmet needs regarding health policy coverage, while community residents were informed of the administrative and political situation of the health care system and services. Such co-learning and mutual understanding enabled multilevel and multifaceted interventions to create an enabling environment for nurturing care.

The last challenge in this study was to minimize the gaps between capacities and needs. Although the community's wishes and expectations were explored in the iterative community assessment, all its needs could not be accommodated in the project. Likewise, although other community health problems or related factors were found, all these could not be addressed in a single project. The goals, benefits, or process of the project should be reasonable, realizable, and reach consensus between the community and research team. A lack of understanding or misunderstanding the project could interrupt the proceedings and strain the relationship between the community and research team. In this study, to avoid potential gaps between the interests and desires of the community and goal and scope of the project, we adopted a theory-guided approach, which provided points to consider within the goal of the project. Transparent and reflective communication was retained throughout the project, which contributed to establishing common ground on various issues. Practically, a memorandum of understanding between each partnership facility and the research team was documented, which outlined the goal and scope of the project and participation of the community.

This study has strengths and significant implications for the long-term sustainability of health promotion practices in disadvantaged communities, where the scarcity of resources and perceived failure of conventional supply-side health promotion programs have been acknowledged. First, this community-driven intervention design included demand-side components that reflected community members' needs and recommendations based on an ecological assessment. This is potentially acceptable, sustainable, and cost-effective, since the participatory approach ensured the intervention components and implementation strategies were culturally sensitive. Second, the power sharing and empowerment process, which facilitates decision-making in community mobilization, transformed the community, instilling higher autonomy and accountability.

Future studies should conduct impact analysis to understand how to replicate or translate this approach in different settings. It is also necessary to understand how to work with governmental organizations to meaningfully contribute to improvements in child health and health equality.

6. Conclusion

This study focused on a theory-guided CBPAR approach to community-driven intervention mapping for child health promotion in migrant communities with few resources. Our study findings showed that the ecological model of an enabling environment for nurturing care informed the intervention mapping process, and the participatory approach ensured culturally acceptable and sustainable intervention mapping alongside community capacity building. The ecological model of an enabling environment for nurturing care provided a perspective on individual and socio-structural contexts and its interdependent influence on nurturing care. Considering socio-structural disadvantages including availability and utilization of health services in migrant communities, an ecological assessment and inter-sectoral approach contributed to addressing underlying issues and complexity. Our ecological model for nurturing care focused on developing an enabling environment to facilitate individual and community empowerment to achieve a greater sense of control over the complex factors related to child health. It was not devoted to providing a single component effort regarding child health at the individual level. Instead, it focused on long-term sustainability of the intervention. It suggests that appropriate framework could bridge the gap between research and community actions in CBPAR.

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CRediT authorship contribution statement

Hyerang Kim: Data curation, Methodology, Writing - original draft. **Soonyoung Shon:** Project administration, Investigation, Data curation, Writing - review & editing. **Hyunsook Shin:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

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